Unit-2 Automate EDA (Exploratory Data Analysis)

602
Data Analytics
using
Python

Contents

2.1 Pandas, NumPy. D-Tale :

Exploratory Data Analysis (EDA) is the process of understanding datasets by summarizing their main characteristics using statistical graphics, plots, and information tables. In Python, three foundational libraries used to support and automate EDA tasks are Pandas, NumPy, and D- Tale. Each of these libraries plays a different role but contributes significantly to making data preparation and understanding more efficient.

Pandas :

2.1.1 Pandas is perhaps the most widely used data manipulation library in Python. It provides powerful data structures such as Series (1D) and DataFrame (2D) which are specifically designed to handle tabular data. With Pandas, you can load data from various file formats like CSV, Excel, JSON, and SQL databases using functions like read_csv(), read_excel(), and read_sql(). Once loaded, the data can be filtered, grouped, sorted, cleaned, and reshaped with ease. Functions like head(), tail(), describe(), info(), value_counts(), and groupby() are particularly useful for EDA. For example, describe() gives statistical summaries of numerical columns, and value_counts() shows the frequency of unique values in a categorical column.

Key EDA Features in Pandas:

Data Loading: read_csv(), read_excel(), read_json()

Data Cleaning: handle missing values (dropna(), fillna()), duplicates

Summarization: info(), describe(), value_counts()

Filtering & Slicing: Select rows/columns using loc, iloc

Grouping and Aggregation: groupby(), agg(), pivot_table()

Visualization Integration: Works well with Matplotlib and Seaborn

NumPy :

NumPy (Numerical Python) complements Pandas by providing efficient operations on large arrays and matrices of numeric data. It supports a wide range of mathematical and statistical operations such as mean, median, standard deviation, dot products, and linear algebra operations. Although Pandas is built on top of NumPy, the latter is especially useful when dealing with low-level numerical data manipulation. For instance, NumPy arrays are faster and more memory-efficient than Python lists for numerical computations.

Role in EDA:

Efficient Computation: Fast array operations (mean, median, std,etc.)

Numerical Summaries: Used under the hood by Pandas and other libraries

Integration: Works smoothly with Pandas, SciPy, and visualization libraries

D-Tale:

D-tale brings interactivity to your data analysis workflow. It combines Pandas with a lightweight Flask server and Dash frontend to provide a web based interface for viewing and editing Pandas DataFrames. You can sort columns, view statistical summaries, filter rows, and even generate charts without writing additional code. The typical workflow involves loading a DataFrame using Pandas and passing it to dtale.show(df), which launches a web interface locally. This makes it easier for analysts and non-technical users to explore the data interactively without diving deep into Python code. D-Tale is a library that lets you view and analyze Pandas DataFrames with an interactive web interface, like Excel + pandas + Plotly.

Features:

1. Launch a web UI for a DataFrame
2. View summary statistics
3. Sort, filter, edit data interactively
4. Visualize distributions (bar charts, histograms, etc.)
5. Detect nulls, outliers, correlations
6. Auto-generates Python code for operations

In summary, Pandas is the core for data manipulation, NumPy enhances numerical performance, and D-Tale provides an interactive interface for data inspection. Together, they lay the groundwork for effective manual or semi-automated EDA.

```
import pandas as pd
import dtale
df =
pd.read_csv("data.csv")
dtale.show(df)
```

2.2 Regression Fundamentals

Introduction to Regression

Regression is a fundamental statistical and machine learning technique used to model the relationship between a dependent variable (target) and one or more independent variables (predictors or features). It allows us to make predictions, understand patterns, and quantify the strength and direction of relationships between variables.

In real life, regression is like predicting someone’s electricity bill based on how many hours they use an air conditioner. As one variable changes, we try to estimate how another will change with it.

2.2.1 Characteristics of Regression

Relationship Analysis What it means:

Regression helps us see how two things are connected. It shows if changing one thing affects the other.

Example:

Suppose you're trying to understand how the number of hours you study affects your exam marks. Regression will help you see if studying more really leads to higher marks.

Analogy:

Think of a remote control and a TV. When you press the volume button (independent variable), the volume changes (dependent variable). Regression tries to understand and measure this type of cause-effect relationship.

Prediction What it means:

Once regression understands the relationship, it can **predict future values**.

Example:

If you studied 4 hours and scored 60 marks, and when you studied 8 hours you scored 90 marks, then regression can help **predict your marks if you study for 6 hours**.

Analogy:

Think of Google Maps. When you enter your starting point and speed, it predicts how long your trip will take. Regression does something similar: **using past patterns to make future predictions**.

Direction and Strength What it means:

Regression tells you **how two things move together**, and **how strongly they're connected**. **There are two directions:**

Positive Relationship:

If one increases, the other increases.

Example: The more hours you sleep, the more energy you have.

Negative Relationship:

If one increases, the other decreases.

Example: The more time you spend on your phone at night, the less sleep you get.

Strength of Relationship:

If studying time and marks are **very closely related**, that's a **strong relationship**.

If there's no pattern, the relationship is **weak or none**. **Analogy**:

Think of holding two magnets:

If they strongly pull toward each other → strong relationship.

If they barely move → weak relationship.

If they push away → negative relationship.

Line of Best Fit What it means:

This is a **line drawn through your data points** that best shows the trend or pattern.

It is not just a random line — it's **mathematically calculated** to reduce the difference between actual values and predicted values.

It gives you a **formula** that you can use to make predictions.

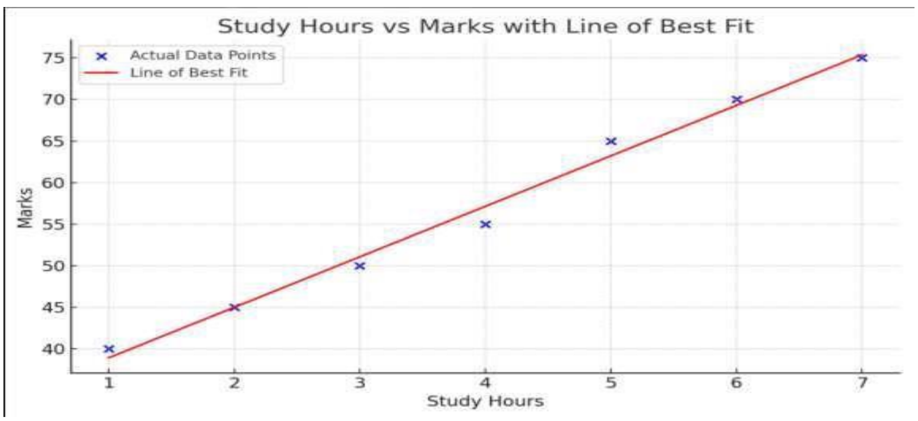
Example:

Imagine you have a bunch of **dots on a graph.**

Each dot shows something like:

1. How many hours you studied
2. What marks you got

Now, you want to draw one straight line that best shows the general pattern of all those dots.



3 Recap in Table Form (characterstics)

Feature	In Simple Words	Real-Life Analogy
Relationship Analysis	Checks if two things are connected	Does more studying lead to better marks?
Prediction	Helps guess the future based on past data	Like predicting rain based on cloud patterns
Direction & Strength	Tells if change is together/opposite, and how strong	Magnets pulling/pushing each other
Line of Best Fit	A line that best shows the trend	A straight path through footprints on sand

2.2.2 Dependent and Independent Variables

Independent Variable (also called Input, Predictor, or Feature):

The variable(s) used to make predictions. This is what you already know.

Example: Hours of study, temperature, years of experience.

Dependent Variable (also called Output or Target):

The variable you want to predict or explain.

Example: Exam score, electricity bill, salary.

Analogy:

Think of making tea:

The amount of sugar and milk you use are independent variables.

The sweetness of the tea is the dependent variable. Changing the input changes the output!

2.2.3 Covariance

Covariance measures how two variables change together:

Positive Covariance: Both variables increase or decrease together.

Negative Covariance: One variable increase while the other decreases.

Analogy:

Imagine two friends who jog together:

If both increase their speed at the same time, the covariance is positive.

If one speeds up and the other slows down, the covariance is negative.

Limitations:

Covariance only tells us the direction of the relationship, not the strength.

Also, the magnitude of covariance is not standardized, so it's hard to compare between datasets.

Correlation

Correlation is a normalized version of covariance that shows both the direction and strength of a linear relationship. - Values range from -1 to +1:

+1: Perfect positive correlation 0: No correlation

-1: Perfect negative correlation

Analogy:

Think of correlation as a friendship scale:

+1: Best friends (do everything together)

0: Strangers (no relation)

-1: Enemies (when one wins, the other loses)

Correlation makes it easier to interpret and compare relationships because it is unitless and scaled.

Summary Table

Concept	Meaning	Analogy
Regression	Predicting a value using known variables	Predicting salary using years of experience
Dependent Variable	The value you want to predict	Exam score
Independent Variable	The value you use to make predictions	Hours studied
Covariance	Direction of joint variation between two variables	Two friends jogging at the same or opposite speeds
Correlation	Direction and strength of linear relationship between variables	Closeness of friendship (-1 to +1)

2.3 Machine Learning Basics:

2.3.1 Concept of Machine Learning

2.3.1.1 Understanding Machine Learning:

What is Machine Learning?

Machine Learning (ML) is a field in computer science where computers learn from data and make decisions or predictions — without being directly programmed for every task.

Instead of writing fixed instructions, we provide examples (called data) and let the machine find patterns on its own.

EXAMPLE:

Think of a child learning to identify animals. First, we show the child pictures of cats and dogs. Then we tell the child, “This is a cat” or “This is a dog".

After seeing many examples, the child starts identifying them correctly, even if it's a new picture.

Machine learning does the same!

We feed a lot of data (examples) into the computer. The computer studies the patterns and then makes predictions on new data.

Real Example:

Email spam filter: Learns which emails are spam based on previous examples.

Netflix suggestions: Learns your taste and suggests shows.

2.3.1.2 Benefits of Machine Learning

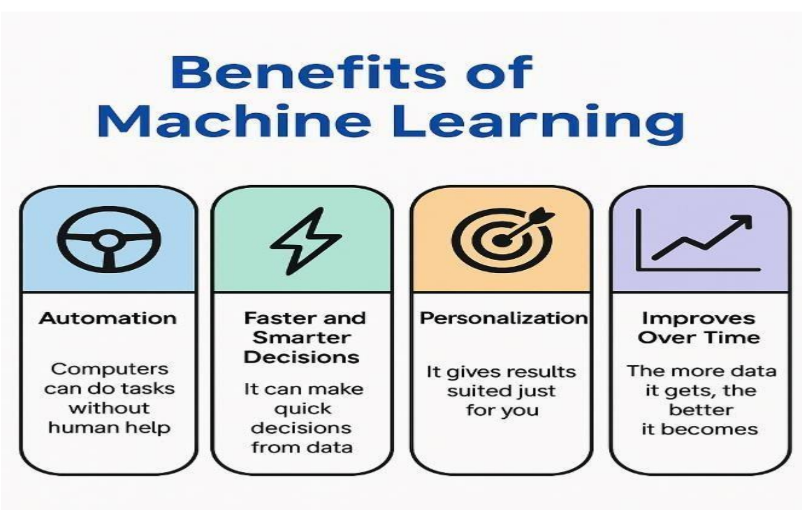
Automation – Computers can do tasks without human help. Ex: Self-driving cars learn how to drive.

Faster and Smarter Decisions – It can make quick decisions from data. Ex: Credit card fraud detection.

Personalization – It gives results suited just for you.

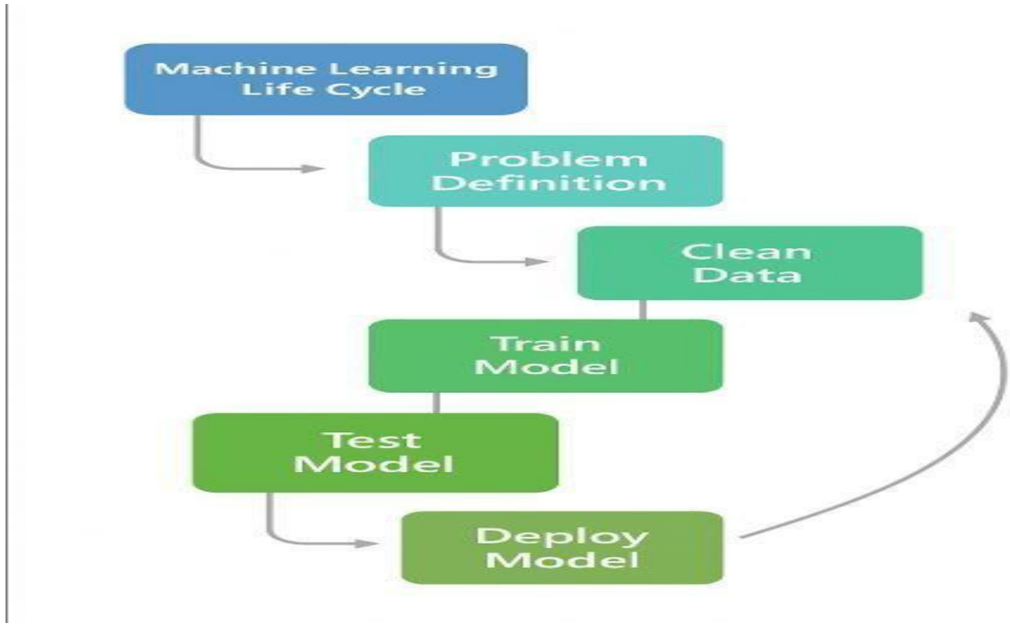
Ex: YouTube recommends videos based on your interests.

Improves Over Time – The more data it gets, the better it becomes. Ex: Voice assistants like Alexa learn your voice over time.
Solves Complex Problems – Like medical diagnosis or language translation.



2.3.1.3 Machine learning Life Cycle

Stages of ML Life Cycle



The Machine Learning (ML) Life Cycle refers to the **step-by-step process** of building, training, testing, and deploying an ML model to solve a real-world problem using data.
Problem Definition What it means:

Define clearly what problem you are trying to solve with ML. Without a clear goal, the project will fail.

Examples:

1. Predict the price of a house
2. Identify whether an email is spam
3. Recommend products to users

Analogy:

It's like a student deciding which subject to study. The goal must be clearly understood before beginning the process.

Data Collection What it means:

Gather the data that is relevant to the problem. This can be done using databases, APIs, surveys, or logs.

Examples:

1. Customer demographics and purchase history

2. Historical stock prices
3. Patient health records

Analogy:

Just like a student collects books and notes to prepare for an exam, ML collects data as its learning material.

Data Cleaning / Preprocessing What it means:

Raw data usually has missing values, errors, duplicates, or inconsistent formats. Cleaning it makes the data usable.

Tasks include:

1. Removing null or duplicate entries
2. Filling missing values with average or most frequent values
3. Converting categorical data into numbers

Analogy:

Like organizing and rewriting messy notes before studying.

Feature Selection and Engineering What it means:

Choose the most relevant variables (features) for the model and sometimes create new ones that help improve model performance.
Examples:

1. Selecting “age” and “income” for a loan model
2. Creating “total_marks” from multiple subject scores

Analogy:

Similar to summarizing your syllabus into key points so you can study better.5

Model Training What it means:

Feed the cleaned and prepared data into a chosen ML algorithm so it can learn from the patterns in the data.

Examples of algorithms:

1. Linear Regression
2. Decision Trees
3. Support Vector Machines

Analogy:

Just like a student study from solved papers and textbooks to prepare for the exam.

Model Testing / Evaluation What it means:

Check how well the model performs using new data that it has never seen before.

Common evaluation metrics:

1. Accuracy
2. Precision and Recall
3. Root Mean Square Error (RMSE)

Analogy:

Like taking a practice test to evaluate how well you’ve studied.

Model Deployment What it means:

Once the model performs well, it is deployed into a real-world environment so that users or systems can use it.

Examples:

1. A chatbot answering customer queries
2. A recommendation engine on an e-commerce site

Analogy:

Like taking the final exam or presenting your project to a panel.

Monitoring and Maintenance What it means:

Even after deployment, the model needs to be checked regularly. If it becomes less accurate due to changing data, it must be retrained or updated.

Examples:

1. Updating the model with new customer data
2. Fixing bugs or improving performance

2.4 Types of Machine Learning

Machine Learning (ML) is the science of enabling computers to learn from data and improve their performance automatically. The learning is broadly categorized into **Supervised Learning, Unsupervised Learning, and Reinforcement Learning** (though reinforcement is not in your unit, it is often mentioned).

2.4.1 Supervised Learning

Definition:

Supervised learning uses **labeled data** to train a model. Each training example has both the input (features) and the correct output (label). The model learns the relationship so it can predict outcomes for unseen data.

How it Works:

1. Provide data with known answers (features + labels).
2. Train the model on this data.
3. Test the model on new data to check if it can generalize.

Examples:

Predicting **house prices** using features like area, location, and number of rooms (Regression).

Classifying **emails as spam or not spam** (Classification).

Detecting **whether a tumor is malignant or benign**.

Advantages:

Produces accurate predictions when sufficient labeled data is available.

Well-established algorithms and evaluation metrics.

Disadvantages:

Requires large, well-labeled datasets (expensive to prepare).

Models may overfit if training data is too small or noisy.

2.4.2 Unsupervised Learning

Definition:

Unsupervised learning deals with **unlabeled data**, where the goal is to discover hidden patterns, structures, or relationships in the data.

How it Works:

1. Input dataset without labels.
2. Algorithm groups similar items or reduces dimensions.
3. Results are clusters, associations, or compressed data.

Applications:

1. Customer segmentation in retail (grouping customers by shopping habits).
2. Market basket analysis (finding items often bought together).
3. Image compression (dimensionality reduction).

Advantages:

1. Works even without labeled data.
2. Helps explore unknown data patterns.

Disadvantages:

1. Results can be subjective and harder to evaluate.
2. No ground truth to compare accuracy.

2.4.3 Applications of Machine Learning

Machine Learning is widely applied across industries because it allows systems to **learn from data, detect patterns, and make decisions or predictions with minimal human intervention**.

1. Healthcare

Disease Prediction & Diagnosis → ML models predict diseases like cancer, diabetes, or heart problems based on medical records, lab results, or imaging.

Medical Imaging → Used in X-rays, MRIs, CT scans for detecting tumors, fractures, or anomalies.

Drug Discovery → Helps in analyzing molecular structures and predicting which compounds may become effective drugs.

Personalized Medicine → Recommends treatments based on individual patient data.

Example: IBM Watson uses ML for oncology research and personalized healthcare recommendations.

2. Finance & Banking

Fraud Detection → ML algorithms detect unusual spending or transaction patterns to prevent fraud.

Credit Scoring & Risk Assessment → Banks use ML to analyze customer financial history and predict loan repayment capability.

Algorithmic Trading → Automated stock trading using ML-based predictions of market trends.

Customer Support → Chatbots and virtual assistants powered by ML handle customer queries.

Example: PayPal uses ML to detect fraudulent transactions in real-time.

3. Retail & E-commerce

Recommendation Systems → Suggests products to customers based on browsing and purchasing history (e.g., Amazon, Flipkart).

Inventory Management → Predicts product demand to optimize stock.

Customer Sentiment Analysis → ML analyzes reviews and feedback to understand customer satisfaction.

Dynamic Pricing → Adjusts product prices automatically based on demand, supply, and competition.

Example: Netflix and Amazon use ML recommendation systems to boost sales and user engagement.

4. Transportation & Autonomous Systems

Self-driving Cars → ML powers object detection, lane detection, traffic sign recognition, and decision-making.

Route Optimization → Predicts traffic congestion and suggests faster routes.

Predictive Maintenance → Identifies potential vehicle breakdowns before they occur.

Example: Tesla's Autopilot system uses ML for real-time driving decisions.

5. Manufacturing & Industry

Quality Control → ML detects defects in production lines through image recognition.

Predictive Maintenance → Predicts equipment failures to reduce downtime.

Supply Chain Optimization → ML forecasts demand and improves logistics planning.

Example: General Electric uses ML for predictive maintenance of industrial machines.

6. Agriculture

Crop Monitoring → ML with sensors and drones detects crop diseases and monitors growth.

Yield Prediction → Predicts crop production based on weather and soil data.

Smart Irrigation → Automates water supply based on soil moisture and weather predictions.

Example: John Deere uses ML in smart farming equipment.

7. Education

Personalized Learning → Adaptive learning platforms customize lessons based on a student's performance.

Automated Grading → ML systems grade assignments, quizzes, and exams.

Learning Analytics → Predicts students at risk of dropping out or failing.

Example: Coursera and Duolingo use ML to personalize learning experiences.

8. Natural Language Processing (NLP) Applications

Virtual Assistants → Siri, Alexa, Google Assistant use ML for speech recognition and response.

Language Translation → Google Translate applies ML for real-time translation.

Chatbots → Customer service bots that learn from conversations.

9. Cybersecurity

Intrusion Detection → Detects unusual behavior in systems to prevent hacking.

Spam & Phishing Detection → Filters malicious emails automatically.

Malware Detection → Identifies harmful software before execution.

10. Entertainment & Media

Content Recommendation → YouTube, Spotify, and Netflix suggest videos/music.

Image & Video Recognition → ML helps in facial recognition and tagging photos.

Game Development → AI opponents in games learn and adapt to player behavior.